1: $(f', b') \leftarrow$ reduced form of the constraint (f_v, b)
▷ b' is a parameter that will be set later
2: **if** f' does not contain a free variable **then**
3: $a \leftarrow$ arbitrary variable of f'
4: $u \leftarrow$ the child of v where α_v satisfy the new constraint $(f'|b')$
▷ $(f'|b')$ is a secondary constraint
5: $\rho_u \leftarrow \rho_v$
▷ The sets of primary constraints, fixed blocks, and marked variables are not changed
6: **else** ▷ f' contains a free variables
7: $a \leftarrow$ arbitrary free variable in f'
8: $u \leftarrow$ a child of v such that $\mathcal{T}_u$ has smaller query size
9: $\rho_u \leftarrow \rho_v, \rho_u[i] \leftarrow$ query x_i ▷ B_i is the block of a
▷ $(f'|b')$ is the primary constraint of the newly fixed block B_i, a is a marked variable
10: Set b' that the constraint $(f'|b')$ is satisfied by all elements of A_u
11: $(M_u|z_u) \leftarrow$ add the constraint $(f'|b')$ to the system $(M_v|z_v)$ and change it to the echelon form by pivoting a
12: $v \leftarrow u$ ▷ New current node